Text Areas : PlainText
Possible Answers :
1

Question Number : 25 Question Id : 69112125 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

If $\alpha = \int_0^{2\sqrt{3}} \log_2\left(x^2+4\right)dx + \int_2^4 \sqrt{2^x-4}\,dx$, then α^2 is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Physics Section A

Section Id :	6911213
Section Number :	3
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	6911213
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 26 Question Id : 69112126 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The dimensional formula of $\frac{1}{2}\epsilon_o E^2$ (ϵ_o = permittivity of vacuum and E = electric field) is $M^aL^bT^c$.

The value of $2a-b+c=$_____.

Options :

69112186. 0

69112187. 1

69112188. -1

69112189. 2

Question Number : 27 Question Id : 69112127 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The diameter of a wire measured by a screw gauge of least count 0.001 cm is 0.08 cm. The length measured by a scale of least count 0.1 cm is 150 cm. When a weight of 100 N is applied to the wire, the extension in length is 0.5 cm, measured by a micrometer of least count 0.001 cm. The error in the measured Young's modulus is $\alpha \times 10^9$ N/m^2. The value of α is _____.
(Ignore the contribution of the load to Young's modulus error calculation)

Options :

69112190. 1.3

69112191. 1.65

69112192. 0.13

69112193. 0.25

Question Number : 28 Question Id : 69112128 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The velocity of a particle is given as $\vec{v} = -x\hat{i} + 2y\hat{j} - z\hat{k}$ m/s. The magnitude of acceleration at point (1, 2, 4) is _____ m/s^2.

Options :

69112194. $\sqrt{6}$

69112195. 9

69112196. $\sqrt{33}$

69112197. 0

Question Number : 29 Question Id : 69112129 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The position of an object having mass 0.1 kg as a function of time t is given as $\vec{r} = \left(10t^2\hat{i} + 5t^3\hat{j}\right)$ m. At $t = 1$ s, which of the following statements are **correct** ?

A. The linear momentum $\vec{p} = \left(2\hat{i} + 1.5\hat{j}\right)$ kg·m/s.

B. The force acting on the object $\vec{F} = \left(2\hat{i} + 3\hat{j}\right)$N.

C. The angular momentum of the object about its origin $\vec{L} = 15\ \hat{k}$ J s.

D. The torque acting on the object about its origin $\vec{\tau} = 20\ \hat{k}$ N m.

Choose the **correct** answer from the options given below :

Options :

69112198. A, B and C only

69112199. B, C and D only

691121100. A, C and D only

691121101. A, B and D only

Question Number : 30 Question Id : 69112130 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A planet (P_1) is moving around the star of mass $2M$ in the orbit of radius R. Another planet (P_2) is moving around another star of mass $4M$ in a orbit of radius $2R$. Ratio of time periods of revolution of P_2 and P_1 is _____.

Options :

691121102. $\frac{1}{2}$

691121103. 2

691121104. 4

691121105. $\frac{1}{4}$

Question Number : 31 Question Id : 69112131 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A particle is rotating in a circular path and at any instant its motion can be described as $\theta = \frac{5t^4}{40} - \frac{t^3}{3}$. The angular acceleration of the particle after 10 seconds is _____ rad/s^2.

Options :

691121106. 150

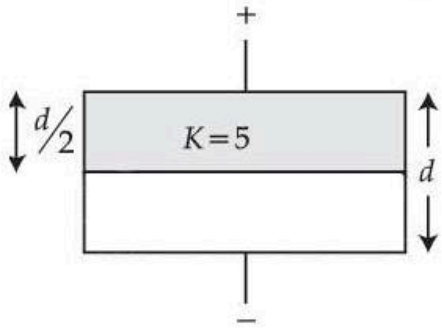
691121107. 120

691121108. 130

691121109. 170

Question Number : 32 Question Id : 69112132 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A parallel plate air capacitor has a capacitance C. When it is half filled as show in figure with a dielectric constant $K=5$, the percentage increase in the capacitance is _____.



Options :

691121110. 33.34

691121111. 66.67

691121112. 200

691121113. 400

Question Number : 33 Question Id : 69112133 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Heat is supplied to a diatomic gas at constant pressure. Then the ratio of $\Delta Q : \Delta U : \Delta W$ is _____.

Options :

691121114. 2 : 3 : 5

691121115. 5 : 3 : 2

691121116. 2 : 5 : 7

691121117. 7 : 5 : 2

Question Number : 34 Question Id : 69112134 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two charged conducting spheres S_1 and S_2 of radii 8 cm and 18 cm are connected to each other by a wire. After equilibrium is established, the ratio of electric fields on S_1 and S_2 spheres are E_{S1} and E_{S2} respectively. The value of $\frac{E_{S1}}{E_{S2}}$ is _____.

Options :

691121118. $\frac{3}{2}$

691121119. $\frac{2}{3}$

691121120. $\frac{4}{9}$

691121121. $\frac{9}{4}$

Question Number : 35 Question Id : 69112135 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The equation of a plane progressive wave is given by $y = 5\cos\pi\left(200\,t - \frac{x}{150}\right)$ where x and y are in cm and t is in second. The velocity of the wave is _____ m/s.

Options :

691121122. 120

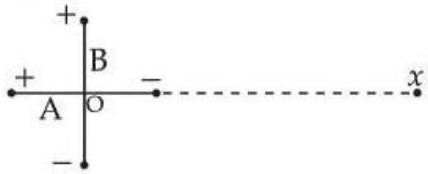
691121123. 150

691121124. 200

691121125. 300

Question Number : 36 Question Id : 69112136 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two short electric dipoles A and B having dipole moment p_1 and p_2 respectively are placed with their axis mutually perpendicular as shown in the figure. The resultant electric field at a point x is making an angle of 60° with the line joining points O and x. The ratio of the dipole moments p_2/p_1 is _____.



Options :

691121126. $\frac{\sqrt{3}}{2}$

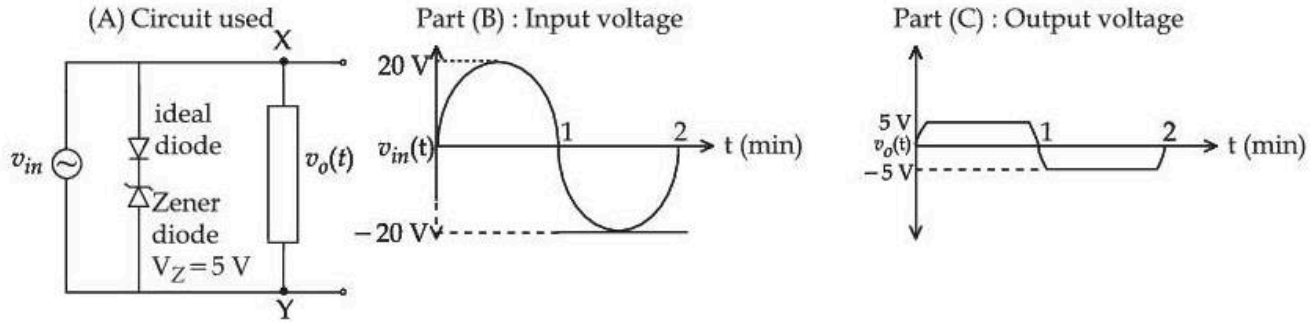
691121127. $2\sqrt{3}$

691121128. $\frac{1}{\sqrt{3}}$

691121129. $\sqrt{3}$

Question Number : 37 Question Id : 69112137 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

For the given circuit (shown in part (A)) the time dependent input voltage $v_{in}(t)$ and corresponding output $v_o(t)$ are shown in part (B) and part (C), respectively. Identify the components that are used in the circuit between points X and Y.



Options :

691121130. 

691121131. 

691121132. 

691121133. 

Question Number : 38 Question Id : 69112138 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

When a coil is placed in a time dependent magnetic field the power dissipated in it is P. The number of turns, area of the coil and radius of the coil wire are N, A and r respectively. For a second coils number of turns, area of the coil and radius of the coil wire are $2N$, $2A$ and $3r$ respectively. When the first coil is replaced with second coil the power dissipated in it is $\sqrt{2}\,\alpha P$. The value of α is _____.

Options :

691121134. 36

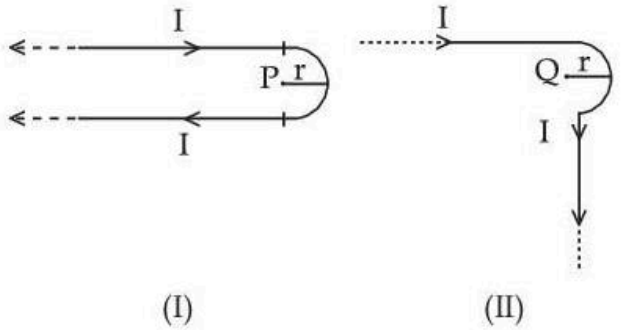
691121135. $128\sqrt{2}$

691121136. 16

691121137. 64

Question Number : 39 Question Id : 69112139 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two identical long current carrying wires are bent into the shapes shown in the following figures. If the magnitude of magnetic fields at the centres P and Q of a semicircular arc are B_1 and B_2 respectively, then the ratio $\frac{B_1}{B_2}$ is _____.



(I) (II)

Options :

691121138. $\frac{2+\pi}{1+\pi}$

691121139. $\frac{1+\pi}{1-\pi}$

691121140. $\frac{2+\pi}{1-\pi}$

691121141. $\frac{1+\pi}{2-\pi}$

Question Number : 40 Question Id : 69112140 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

For a thin symmetric prism made of glass (refractive index 1.5), the ratio of incident angle and minimum deviation will be _____.

Options :

691121142. 3 : 4

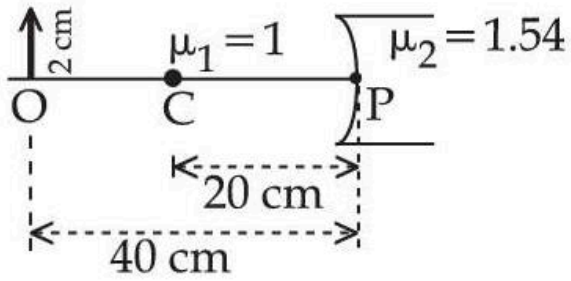
691121143. 3 : 2

691121144. 2 : 1

691121145. 1 : 2

Question Number : 41 Question Id : 69112141 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Refer the figure given below. μ_1 and μ_2 are refractive indices of air and lens material. The height of image will be _____ cm.



Options :

691121146. 1

691121147. 0.5

691121148. 1.2

691121149. 0.25

Question Number : 42 Question Id : 69112142 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

For a certain metal, when monochromatic light of wavelength λ is incident, the stopping potential for photoelectrons is $3V_o$. When the same metal is illuminated by light of wavelength 2λ, then the stopping potential becomes V_o. The threshold wavelength for photoelectric emission for the given metal is αλ. The value of α is _____.

Options :

691121150. 1

691121151. 4

691121152. 2

691121153. 3

Question Number : 43 Question Id : 69112143 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

An electromagnetic wave travelling in x-direction is described by field equation $E_y = 300 \sin \omega \left(t - \frac{x}{c} \right)$. If the electron is restricted to move in y-direction only with speed of 1.5×10^6 m/s then ratio of maximum electric and magnetic forces acting on the electron is _____.

Options :

691121154. 200

691121155. 150

691121156. 400

691121157. 300

Question Number : 44 Question Id : 69112144 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Angular momentum of an electron in a hydrogen atom is $\frac{3h}{\pi}$, then the energy of the electron is _____ eV.

Options :

691121158. -1.51

691121159. -0.85

691121160. -0.38

691121161. -0.28

Question Number : 45 Question Id : 69112145 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A liquid drop of diameter 2 mm breaks into 512 droplets. The change in surface energy is $\alpha \times 10^{-6}$ J. The value of α is _____. (Take surface tension of liquid = 0.08 N/m)

Options :

691121162. 10

691121163. 7

691121164. 8

691121165. 11

Physics Section B

Section Id :	6911214
Section Number :	4
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	6911214
Question Shuffling Allowed :	Yes

Is Section Default? : No

Question Number : 46 Question Id : 69112146 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

In single slit diffraction pattern, the wavelength of light used is 628 nm and slit width is 0.2 mm, the angular width of central maximum is $\alpha \times 10^{-2}$ degrees. The value of α is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 47 Question Id : 69112147 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A vessel contains 0.15 m^3 of a gas at pressure 8 bar and temperature 140 °C with $c_p = 3R$ and $c_v = 2R$. It is expanded adiabatically till pressure falls to 1 bar. The work done during this process is _____ k J. (R is gas constant)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 48 Question Id : 69112148 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

1 μC charge moving with velocity $\vec{v} = \left(\hat{i} - 2\hat{j} + 3\hat{k}\right)$ m/s in the region of magnetic field $\vec{B} = \left(2\hat{i} + 3\hat{j} - 5\hat{k}\right)$ T. The magnitude of force acting on it is $\sqrt{\alpha} \times 10^{-6}$ N. The value of α is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 49 Question Id : 69112149 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A uniform wire of length l of weight w is suspended from the roof with a weight of W at the other end. The stress in the wire at $\frac{l}{3}$ distance from the top is $\left(\frac{W}{A} + \frac{2}{\gamma}\frac{w}{A}\right)$, where, A is the cross sectional area of the wire. The value of γ is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

Question Number : 50 Question Id : 69112150 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A tub is filled with water and a wooden cube 10 cm × 10 cm × 10 cm is placed in the water. The wooden cube is found to float on the water with a part of it submerged in water. When a metal coin is placed on the wooden cube, the submerged part is increased by 3.87 cm. The mass of the metal coin is _____ gram.

(Take water density as 1 g/cm^3 and density of wood as 0.4 g/cm^3)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Chemistry Section A

Section Id :	6911215
Section Number :	5
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	6911215
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 51 Question Id : 69112151 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The mass of iron converted into Fe_3O_4 by the action of 18 g of steam is :

(Given : Molar mass of H, O and Fe are 1, 16 and 56 g mol^{-1} respectively)

Assume iron is present in excess :

Options :

691121171. 2.1 g

691121172. 4.2 g

691121173. 21 g

691121174. 42 g

Question Number : 52 Question Id : 69112152 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical